



Exploring the therapeutic efficacy of Bai-Shao in mitigating comorbid epileptic seizures and cognitive impairment via inflammatory signaling pathways: insights from in silico and in vivo studies

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Abstract

Bai-Shao (BS) ingredients capable of crossing the blood–brain barrier (BBB) were identified and analyzed for their main gene targets. Genes associated with epileptic seizures (ES), cognitive impairment (CI), and inflammation were retrieved, with common targets between BS and these conditions determined via Venn analysis. Protein–protein interaction (PPI) analysis identified the top 10 key genes, whose correlations with apolipoprotein E (APOE) and methylenetetrahydrofolate reductase (MTHFR) were assessed. Functional pathways were explored using Gene Ontology (GO) and the Kyoto Encyclopedia of Genes and Genomes (KEGG), while molecular docking evaluated interactions between BS compounds and target proteins. Findings were validated in a pentylenetetrazol (PTZ)-induced mouse model. BS contains β -sitosterol and betulinic acid, both BBB-permeable, with 277 shared ES/CI-related targets identified. The top 10 genes correlated with APOE and MTHFR, and pathway analysis highlighted interleukin-17 (IL-17) and tumor necrosis factor (TNF) signaling, involving FOS proto-oncogene (FOS), jun proto-oncogene (JUN), caspase-3 (CASP3), phosphoinositide 3-kinase (PI3K), protein kinase B (AKT), and cyclooxygenase-2 (COX2). Behavioral tests, including the Open Field Test and Morris Water Maze, indicated cognitive and motor improvements in PTZ-treated mice. Molecular analysis revealed hippocampal regulation of key genes and altered TNF- α (TNF α), interleukin-6 (IL-6), and interleukin-1 beta (IL-1 β) levels. In conclusion, BS demonstrates therapeutic potential for comorbid ES and CI by modulating inflammatory pathways and cell survival mechanisms, supporting its role in neurological research.

Keywords Bai-Shao · Epilepsy · Cognitive dysfunction · Network pharmacology · Inflammation

Introduction

Epilepsy is a debilitating chronic brain disorder characterized by recurrent, unprovoked seizures (Reddy et al. 2018). This condition affects a significant global population, with more than 70 million individuals of diverse demographics being impacted by the disease (Cao et al. 2024). However, it is important to note that epilepsy extends beyond the manifestation of epileptic seizures (ES), as patients often encounter a range of psychological, cognitive, and social changes (Elger et al. 2004; Boylan et al. 2004). Notably, these

comorbid conditions often pose a greater threat to patients than the seizures they experience and significantly deteriorate their overall quality of life (Man Chen et al. 2021; Mula and Sander 2007). The co-occurrence of cognitive impairment (CI) in individuals with ES encompasses a range of challenges, such as impairments in attention, memory, and cognitive processing abilities (Holmes 2015).

Epilepsy individuals experience a significantly elevated prevalence of CIs, which have a detrimental impact on their educational advancement and lifelong achievements (Berg et al. 2012). Despite the widely acknowledged and prevalent nature of behavioral comorbidities in epilepsy, the underlying causes of these conditions remain poorly understood (Carron et al. 2019). Furthermore, exploring the potential shared mechanisms between epilepsy and its comorbidities holds significant scientific interest. From a fundamental research standpoint, investigating the role of inflammation and activation of astrocytes and microglia in the mechanisms

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that contribute to heightened epileptogenesis and excitability is well-established (W. Li et al. 2023). Moreover, neuroinflammation plays a pivotal role in the pathogenesis of numerous neurological disorders associated with CI (Qi-yue Zhang et al. 2024; Junjie Chen et al. 2025).

The current antiepileptic drugs (AEDs) have limited effectiveness in managing epilepsy. They can only alleviate symptoms and manage seizures in approximately 70% of individuals. Furthermore, none of the currently available AEDs have proven capable of preventing spontaneous recurrent seizures or adequately addressing the psychiatric and neurological comorbidities linked to epilepsy (Leo et al. 2019). Consequently, managing seizures alongside associated neuropsychiatric conditions poses a considerable therapeutic challenge. Hence, it is crucial to explore novel pharmaceutical solutions to address these comorbidities.

Traditional Chinese Medicine (TCM) often utilizes herbal components to treat neuropsychiatric disorders due to their anti-inflammatory properties, which come with fewer side effects and lower toxicity compared to other treatments (Hongli Li et al. 2022a; Hui Zhang et al. 2023; Hui et al. 2024). TCM have demonstrated the ability to modulate neuroinflammation and oxidative stress through their bioactive ingredients (Tripathi et al. 2024). The White Peony (*Paeonia lactiflora* Pall.) root, known as Bai-Shao (BS) or *Paeoniae Radix Alba* (白芍), is a well-known example of TCM (Du et al. 2020). Pharmacological investigations have reported the anti-inflammatory, immune-regulatory, and neuroprotective characteristics of BS (Chang et al. 2015).

Targeted treatment of neurodegenerative diseases can enhance the effectiveness of therapy (Ramakrishna et al. 2023; Tripathi et al. 2019; Srivastava et al. 2019; Rai et al. 2020). Network pharmacology is an emerging method in drug development that utilizes the principles of systems biology to explore the potential effects of drugs, their targets, and their pharmacological functions. It involves analyzing the networks within biological systems and identifying specific signaling components for the purpose of modifying and designing drugs (Saihong Zhao et al. 2023; Qin et al. 2023; Ramakrishna et al. 2024). The use of network pharmacology approach in TCM studies has gained significant popularity, as it provides a crucial framework for comprehending the pharmacological mechanisms of these drugs (Zheng et al. 2022; Cheng et al. 2024). However, it is essential to note that network pharmacology predictions require further validation, such as through the implementation of molecular docking, which is a computational method designed to assess the binding affinity between receptors and drugs (Agu et al. 2023).

The study focused on evaluating the therapeutic effects of BS extract on the co-occurrence of epileptic seizures and cognitive impairment (ES/CI). To accomplish this, we utilized a comprehensive approach that combined network

pharmacology, protein–protein interaction (PPI) network, molecular docking analysis, and validation through animal studies. The main goals of this research were to confirm the key bioactive components of BS extract, identify potential target genes, and clarify the mechanisms that underlie its pharmacological effects.

Methods and materials

In silico study

Prediction of potential bioactive ingredients and targets genes

To identify the active ingredients in BS, we conducted a comprehensive investigation utilizing multiple databases. Our initial search was performed in the Traditional Chinese Medicine Systems Pharmacology (TCMSP) database (available at: <https://tcmsp-e.com/>). We applied specific filter criteria, including an oral bioavailability (OB) of 30% or higher, a drug-likeness (DL) score of at least 0.18, and a blood–brain barrier (BBB) permeability value of no less than -0.3 to identify a list of suitable active ingredients in BS that have the potential to cross the BBB and exert therapeutic efficacy in neurological conditions (H. Y. Xu et al. 2019). Each collected ingredient was entered into the PubChem database to retrieve their CAS number and canonical SMILES (Dashti et al. 2019). To further evaluate the potential of these compounds as active ingredients, we used the SwissADME platform (available at: <https://www.swissadme.ch/>) (Daina et al. 2017). We used the canonical SMILES of the compounds as input and assessed their characteristics, including gastrointestinal absorption (GI absorption) and drug-like properties based on various criteria. Compounds with high GI absorption and positive results in at least two of the five classes (Ghose, Lipinski, Egan, Veber, and Muegge) were considered potential active ingredients (Zhang et al. 2021). Furthermore, ProTox 3.0 was used to assess the potential toxicity of the bioactive ingredients (Banerjee et al. 2018).

Target gene prediction for BS

To determine the targets associated with the bioactive ingredients, we used the SwissTargetPrediction database (available at: <https://www.swisstargetprediction.ch/>), SuperPred (available at: <https://prediction.charite.de/>), and "Find my Compound's Target" in BindingDB (available at: <https://www.bindingdb.org/rwd/bind/index.jsp/>). For SwissTargetPrediction database, we inputted the canonical SMILES of the bioactive ingredients and obtained the related target genes (Daina et al. 2019). To aid researchers

in understanding a compound's mechanism of action, we utilized the BindingDB (available at: <https://www.bindingdb.org/rwd/bind/index.jsp/>), which predicts potential targets for compounds. (T. Liu et al. 2007). Additionally, SuperPred (available at: <https://prediction.charite.de/>) is another online platform to forecast target genes of ligands based on their structure (Nickel et al. 2014). Finally, to ensure uniformity in gene symbols, we referred to the UniProt platform (available at: <https://www.UniProt.org>) to convert the gene symbols sourced from the SwissTargetPrediction database (Ahn et al. 2019).

Related targets of cognitive impairment, epilepsy, and inflammatory diseases

We collected targets of CI, epilepsy, and inflammatory diseases via different databases to prepare for subsequent analysis. The keywords “Mild cognitive disorders or cognition disorders”, “epilepsy”, and “Inflammation or inflammatory disorder” were used in DisGeNET (available at: <https://www.disgenet.org/>) (Piñero et al. 2016), GeneCards (available at: <https://www.genecards.org/>, ver. 4.9.0) (Stelzer et al. 2016), and Comparative Toxicogenomics Database (available at: CTD; <http://ctdbase.org/>) (Davis et al. 2021). Furthermore, all target genes were standardized using the UniProt platform (UniProt: the universal protein knowledge-base in 2021, 2021).

Prediction of shared genes

To identify the shared targets of BS with CI, ES, and inflammatory diseases, we utilized a Venn diagram created using Venny 2.1.0 0 (available at: <http://bioinfogp.cnb.csic.es/tools/Venny/index.html/>). Venny is an effective tool for visualizing overlapping sets, allowing us to clearly illustrate the common targets between these conditions. The resulting Venn diagram highlights the intersections, providing valuable insights into the relationships among these diseases.

Protein–protein interaction

After identifying the BS-related and ES/CI-related targets, we determined the intersecting genes and constructed a protein–protein interaction (PPI) network using the STITCH 5.0 database (available at <http://stitch.embl.de/>), filtering exclusively for Homo sapiens interactions (Schneeweiss et al. 2015). The resulting network data was downloaded in TSV format and visualized in Cytoscape (version 3.9.0). To rank the key genes within the network, we employed the CytoNCA plugin to perform a detailed topology analysis. Three centrality measures were used: degree centrality, which assesses the number of direct interactions a node has; betweenness centrality, which evaluates a node's role in

mediating interactions between other nodes; and closeness centrality, which indicates how quickly a node can interact with all other nodes in the network. Based on these parameters, the top 10 key genes with the highest centrality scores were selected as core targets for further investigation.

Correlation analysis between key targets and key genes of AD

To confirm the relationship between key genes and major causative Apolipoprotein E polymorphism (APOE) and methylenetetrahydrofolate reductase (MTHFR) TT genotype, as well-recognized genetic risk factors for mild CI correlation analysis (Duan et al. 2023), was conducted using GEPIA <http://gepia.cancer-pku.cn/index.html> (Zefang Tang et al. 2017). APOE and MTHFR were chosen for correlation analysis based on their well-documented associations with cognitive function and neuroinflammation existing literature. Specifically, APOE is recognized for its significant impact on Alzheimer's disease (AD) risk and cognitive decline (Bunce et al. 2004; Martins et al. 2005), while MTHFR has been implicated in folate metabolism and its relation to neurodevelopmental and neurodegenerative disorders (Piras et al. 2020; Barati et al. 2022).

GO and KEGG pathway

Functional pathways associated with the intersecting target genes were identified using Gene Ontology (GO) (Tiancheng Ma et al. 2021) and Kyoto Encyclopedia of Genes and Genomes (KEGG) (Ge et al. 2020) enrichment analyses, conducted through the ShinyGO 0.77 online platform (available at <http://bioinformatics.sdstate.edu/go/>). For the GO analysis, target genes were categorized into three dimensions: molecular function (MF), biological process (BP), and cellular component (CC), with a significance threshold set at $p < 0.05$. Only the top 10 most enriched GO terms in each category were retained, and the results were visualized using bar plots to highlight the most relevant functional annotations. In parallel, KEGG pathway enrichment analysis was performed using stringent criteria of $p < 0.05$ and a false discovery rate (FDR) < 0.05 . The significantly enriched KEGG pathways were illustrated using dot plots. Key targets were further prioritized based on their prominent representation in these enriched GO and KEGG pathways, ensuring that only genes playing crucial roles in the identified functional networks were selected for subsequent analyses.

Molecular docking verification

We selected the top five proteins identified from the PPI analysis as receptors for docking studies with the bioactive components derived from BS, which served as ligands.

The three-dimensional structures of the receptor proteins were obtained from the Protein Data Bank (PDB; available at: <https://www.rcsb.org/>) (Xiaofei Ma et al. 2019). Specifically, we downloaded the crystal structures for the Jun-Fos-DNA complex (PDB ID: FOS1), MAPK1 (PDB ID: 6G54), HSP90 AA1 (PDB ID: 1UY9), and PIK3 CA (PDB ID: 2RD0). The three-dimensional structures of the bioactive components were retrieved from PubChem (available at: <https://pubchem.ncbi.nlm.nih.gov/>) in “sdf” format. To facilitate docking, we employed CB-Dock2 (available at: <https://cadd.labshare.cn/cb-dock2/php/index.php/>), which automatically identifies binding sites on the receptor, calculates their dimensions and central coordinates, and adjusts the docking box size accordingly. This automated approach ensured the accurate positioning of the ligand within the receptor’s binding pocket. The docking simulations were executed using the integrated AutoDock Vina algorithm, and the outcomes were visualized to assess the interactions between the core receptors and the bioactive ingredients. For the visual representation of the docking results, the ligands were displayed in a “ball and stick” format, while the receptors were rendered in both “Backbone” and “Surface” formats. The coloration scheme was applied such that the ligand’s color reflected its elemental composition, and the receptor’s color was assigned based on hydrophobicity (Nie et al. 2022). This detailed visualization facilitated the interpretation of binding interactions and the assessment of potential binding affinity and stability between the bioactive components and the target proteins.

In vivo Study

Extract preparation

The extract was obtained following previous studies (Xiao Li et al. 2021). In summary, BS was soaked in 70% ethanol (1:8, w/v) for 2 h and then extracted twice with 70% ethanol for another 2 h. The extract was subsequently concentrated under vacuum, lyophilized into a powder, and stored at -80°C , with a drug-extract ratio of 15.1%.

Animals

In this study, Male C57BL/6 J mice (6–8 weeks old, 20–25 g) were used exclusively to minimize variability from hormonal fluctuations in females, which can influence seizure susceptibility and cognitive outcomes (Velíšková and DeSantis 2013). They were kept in ventilated polypropylene cages under controlled laboratory conditions, maintained at a temperature of $22 \pm 2^{\circ}\text{C}$, with a 12-h light/dark cycle, for a one-week acclimatization period prior to the commencement of the experiment. During this acclimation phase, the animals had unrestricted access to a standard diet and fresh

water. The experimental procedures were conducted following the approval of the Research Ethics Committee on the Care and Use of Laboratory Animals at Shandong Provincial Third Hospital, Shandong University (No: SDZY220028), adhering to the guidelines established by the NIH for use of laboratory animals. Every possible measure was taken to minimize the suffering of the rodents throughout the course of the study.

Study design

In this study, 72 mice were randomly and blindly assigned to six experimental groups ($n = 12$ per group) as follows: Group I (CON group): Received the vehicle for BS per oral (p.o.) and the vehicle for pentylenetetrazole (PTZ) intraperitoneally (i.p.); Group II (PTZ group): PTZ-treated animals received the vehicle for BS (p.o.); Group III (PTZ + VLP group): PTZ-treated animals received valproate (VLP); and Groups IV, V, and VI (PTZ + BS1, PTZ + BS2, and PTZ + BS3): PTZ-treated animals ($n = 10$ per group) received BS extract at doses of 100, 200, and 400 mg/kg, respectively. The BS doses were selected based on previous studies to ensure biological relevance and efficacy (Hyun Ok Yang et al. 2004; Wenjing Ma et al. 2024). PTZ (35 mg/kg; Sigma-Aldrich) was dissolved in sterile isotonic saline (vehicle). Animals in the treatment groups received the respective doses of BS extract or the vehicle (0.5% DMSO) via oral gavage 1 h before PTZ injection. VLP (100 mg/kg, i.p.; Sigma-Aldrich) was used as a positive control anticonvulsant drug and was administered 1 h before PTZ injection. To minimize potential confounding factors, randomization was performed using a computer-generated sequence, and all investigators were blinded to treatment allocation throughout the study. Additionally, environmental conditions (temperature, humidity, and light–dark cycle) were carefully controlled, and all animals underwent the same handling procedures to reduce variability.

Establishment of Epilepsy Model

Epileptic seizures were induced in mice as previously described (Zhu et al. 2020). Briefly, animals received an intraperitoneal injection of PTZ (35 mg/kg; Sigma-Aldrich) every two days, for a total of 11 injections over a 21-day period. Mice that experienced more than three consecutive stage 4 seizures were classified as fully kindled. Following each PTZ injection, animals were continuously monitored for 45 min, and seizure intensity was assessed using a modified Racine scale. Seizure severity was categorized as follows: Grade 0 indicated no response, Grade 1 involved ear and facial twitching, and Grade 2 was characterized by myoclonic jerks and rearing. Grade 3 included myoclonic jerks with rearing, while Grade 4 exhibited wild running,

jumping, and falling onto the side. Grade 5 represented generalized tonic–clonic seizures with loss of posture, and Grade 6 indicated death. In addition to seizure grading, data on the number of seizure episodes, their onset latency, and total duration were meticulously recorded throughout the study. To assess potential off-target effects of PTZ and treatment interventions, animals were closely observed for abnormal behaviors unrelated to seizure activity, such as changes in locomotion, feeding patterns, and body weight. No significant off-target effects were detected, further supporting the specificity of the PTZ-induced epilepsy model.

Behavioral tests

The Open Field Test (OFT) serves as a tool for assessing anxiety levels, spontaneous activity, and exploratory behavior in experimental animals, while the Morris Water Maze (MWM) evaluates spatial learning and memory capabilities (R. Zhang et al. 2015). The sample size of $n = 12$ per group for behavioral assessments was determined based on previous studies utilizing the PTZ-kindling model. This choice aligns with standard practices in rodent epilepsy research, balancing statistical reliability with ethical considerations to minimize animal use. A blind researcher in a controlled, quiet environment maintained at 22 ± 2 °C to ensure consistency across all assessments conducted behavioral evaluations. During both the OFT and MWM, a video camera was positioned directly above the center of the testing area, enabling real-time monitoring and recording of the animals' behaviors throughout the experiments. Automated video tracking software was utilized to analyze movement patterns, reducing observer bias and minimizing variability in the recorded outcomes, as part of the broader strategy to ensure reproducible and reliable behavioral data.

OFT: In this study, the OFT was conducted as described in a previous publication (R. Zhang et al. 2015). Each animal was placed in the open field arena (50 cm in height and 120 cm in diameter) that was divided into 25 transparent regions, covering the central, peripheral, and intermediate areas of the arena. The animal was given the opportunity to navigate the arena freely for a duration of 3 min. During this time, the researchers documented the duration the mouse remained motionless in the central zone (referred to as latency) as well as the frequency with which the mouse crossed the grid lines.

MWM: The day following the OFT, we conducted the MWM through a modified five-day procedure specifically designed for mice (R. Zhang et al. 2015). The experimental setup involved a 120 cm diameter plastic pool filled with water to a depth of 31 cm, maintained at a temperature of 23–25 °C. For analysis, the interior of the pool was effectively segmented into four quadrants. In the second quadrant, an escape platform measuring 8 cm in diameter was

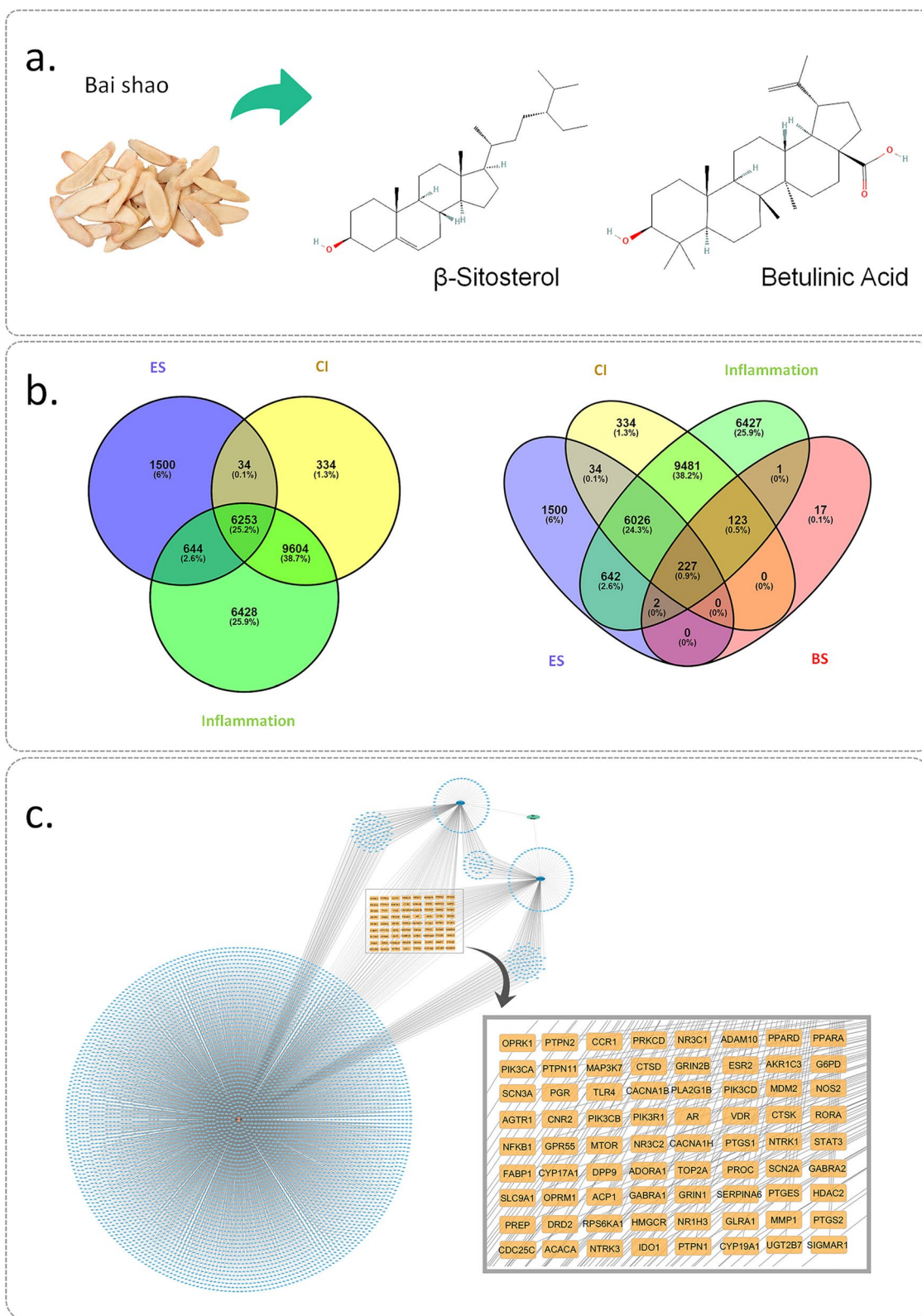
concealed just beneath the water's surface, positioned 1 cm below. The testing protocol comprised four days of escape trials, followed by a single probe trial on the fifth day. During the escape trials, each mouse underwent four trials daily with an interval of 60 s between each trial. Initially, the animal was placed in the water at the center of each quadrant's edge, oriented towards the wall of the pool. The duration it took for each mouse to find the hidden platform, referred to as escape latency, was carefully recorded. If a mouse did not locate the platform within 1 min, it was gently taken out of the water and positioned on the platform for 10 s. On the fifth day, the platform was removed to carry out the probe trial, allowing the animals to explore freely. The frequency with which each mouse crossed the center of quadrant 2 (termed passing times) was noted within a 60-s timeframe. Following each trial, the animals were dried off and then placed back in their cages. The data gathered from both the escape latency during the escape trials and the passing times recorded in the probe trial were subsequently analysed to gain insights into the mice's spatial learning and memory capabilities.

Tissue collection and preparation

For qRT-PCR and ELISA, a sample size of six mice per group was utilized for each experiment. The mice were anesthetized through an intraperitoneal injection of a solution that included 10% chloral hydrate (400 mg/kg) and sodium pentobarbital (100 mg/kg). Following the administration of anesthesia, the brains were swiftly excised from the skulls. The hippocampal tissues were carefully isolated from each brain on ice and then stored in a freezer at -80 °C.

qRT-PCR

Fresh hippocampal tissues were used for extraction of total RNA using Trizol reagent (Invitrogen). The concentrations of RNA assessed by measuring absorbance at a wavelength of 260 nm. To synthesize first-strand complementary DNA (cDNA), we employed a Reverse Transcription System (Qiagen). qRT-PCR was performed using Maxima SYBR Green dye (Fermentas), utilizing an Eppendorf thermocycler (Hamburg, Germany). The reaction mixture included Maxima SYBR Green Master Mix, the cDNA sample, and both forward and reverse primers. The specific primers utilized were as follows: Jun forward, GCGTGAAGAGSGGCTTG TG, and R, CAGTATCAAAGTTGAGAGGAAGC; Fos F, GTTGTCTCCTCGGACTCTA, and R, AATCTCAATAAT GAACCACACA; Cyclooxygenase-2 (COX-2) F, CCTGGC TGCTTCGCTTTACA, and reverse, TGTGGAACCCTT TCTTTGTC; Caspase-3 (Casp3) F, AGGCAAGCAAGA TTATCTGCA, and R, AGCACGCTACAGTGATCATAA; PI3 K F, GCACATCCGTGACTGCACAT, and R, CAG



CGCACTGTCATGGTATG; AKT F, TAGCCATTGTGA AGGAGGGC, and R, CCTAGGCGCGTCTTCTTGGA; and GAPDH F, GAAGTTACTCGTCTGGTCCTCA, and

reverse, TGTGCCAGGTGTTCTTCATC. The levels of the target genes were normalized against GAPDH, which served as a housekeeping gene. The analysis of gene expression

Fig. 1 "Bioactive Ingredients and Target Identification of Bai-Shao (BS) in ES/CI. **a** Chemical structures of BS's bioactive compounds, β -sitosterol and betulinic acid, identified via TCMSP database screening for oral bioavailability ($OB \geq 30\%$), drug-likeness ($DL \geq 0.18$), and blood–brain barrier permeability ($BBB \geq -0.3$). **b** Venn diagrams showing overlapping targets: (A) 6253 common targets among epilepsy (ES), cognitive impairment (CI), and inflammation from DisGeNET, GeneCards, and CTD databases; (B) 277 shared targets between BS bioactive components and ES/CI-related genes. **c** Drug–target–disease interaction network constructed in Cytoscape, comprising 6400 nodes (2 BS ingredients, 277 targets) and 6738 edges, illustrating the interconnectedness of BS compounds with ES/CI targets

was performed using the $2^{(-\Delta\Delta Ct)}$ method, allowing for quantification relative to the control.

ELISA

The levels of hippocampal proteins IL-6, TNF- α , and IL-1 β were determined through ELISA. The ELISA kits for these proteins were sourced from Bioassay Technology Laboratory and utilized following the instructions provided by the manufacturer.

Statistical analysis

Statistical evaluations were performed using GraphPad Prism (Version 8.0, GraphPad Software, San Diego, CA, USA), with all results expressed as mean $\pm$ standard error of the mean (SEM). Behavioral and molecular data were analyzed using one-way analysis of variance (ANOVA) to assess differences across experimental groups (CON, PTZ, PTZ + VLP, and PTZ + BS at 100, 200, and 400 mg/kg). For pairwise group comparisons, post hoc analyses were conducted using Tukey's multiple comparison test. A significance level of $p < 0.05$ was established to determine statistically significant differences between groups, with no additional adjustments for multiple comparisons due to the hypothesis-driven design and limited number of planned comparisons. Significant results are denoted in figures as "a" (vs. CON), "b" (vs. PTZ), and "c" (between treatment groups), as indicated in the respective legends.

Results

Findings from In Silico Study

Potential bioactive components of Bai Shao

Thirty-nine potential bioactive compounds were identified in BS analysis. Among these, two active ingredients, representing 5.12%, were determined to satisfy the established screening criteria. The identified active compounds are beta-Sitosterol and Betulinic Acid. Structural data for all

active components were sourced from PubChem (Fig. 1a). In addition, ADME profiles for each active ingredient were collected (Tables 1 and 2). β -sitosterol (bSS, PubChemID: 222,284, Mol ID: MOL000358) and betulinic acid (BA, PubChemID: 64,971, Mol ID: MOL000211) are two bioactive compounds with distinct molecular properties. β -sitosterol has a molecular weight of 414.79 g/mol, one hydrogen bond acceptor (Hacc), one hydrogen bond donor (Hdon), and six rotatable bonds (Rbon). Its topological polar surface area (TPSA) is 20.23 Å, and it violates the Lipinski rule of five with a MLOGP value greater than 4.15. The GIA score is -2.20 , and the Log Kp (cm/s) value is 0.55. On the other hand, BA has a molecular weight of 456.70 g/mol, three Hacc, two Hdon, and two Rbon. It has a TPSA of 57.53 Å and also violates the Lipinski rule of five. The GIA score is -3.26 , and the Log Kp (cm/s) value is 0.85. The results indicated that the LD50 values for bSS and BA were found to be 890 mg/kg and 2610 mg/kg, respectively. The possible toxicological characteristics of each component was summarized in Table 3. In terms of organ toxicity, bSS was inactive in causing hepatotoxicity, neurotoxicity, and nephrotoxicity, while showing activity in terms of respiratory toxicity and cardiotoxicity. Similarly, BA was inactive in causing hepatotoxicity, neurotoxicity, and nephrotoxicity, while exhibiting activity in respiratory toxicity and cardiotoxicity. Both compounds showed activity in terms of carcinogenicity and immunotoxicity, while being inactive in mutagenicity and cytotoxicity.

Common target prediction

Upon the extraction and removal of duplicate genes from diverse disease online platforms, a thorough analysis exposed a total of 8,431 targets related with ES, 16,225 targets associated with CI, and 22,929 targets associated with inflammation. Of these, 6,253 targets were shared among ES, CI, and inflammation (Fig. 1b). Furthermore, the investigation identified 370 targets linked to BS (Bioactive Substance) from two specific ingredients. Notably, 277 targets were found to be shared between BS and the aforementioned diseases (Fig. 1b).

Interaction network

To explore the intricate relationship between the bioactive ingredients of BS and their potential impacts on ES/CI, we implemented a thorough methodology. Following the identification of targets linked to BS and its associated ingredients, we developed a drug–target–disease network, presented by Cytoscape (Fig. 1c). This network consisted of 6400 nodes and 6738 edges, incorporating two unique ingredients from BS along with 277 targets shared with various diseases. The connections between these nodes, represented by the 6738

Table 1 Bioactive compounds of Bai Shao

PubChemID	Mol ID	Molecule Name	Formula	OB (%)	BBB	DL	BBB Permeant
222,284	MOL000358	β -sitosterol	C29H50O	47.72	0.99	0.75	No
64,971	MOL000211	Betulinic Acid	C30H48O3	39.27	0.22	0.78	No

OB Oral Bioavailability, BBB Blood–Brain Barrier, DL Drug-Likeness

Table 2 Physicochemical properties of components

PubChemID	MW (g/mol)	Hacc	Hdon	Rbon	TPSA (Å)	Lipinski	GIA	Log Kp (cm/s)	BS
222,284	414.79	1	1	6	20.23	1 violation: MLOGP > 4.15	Low	– 2.20	0.55
64,971	456.70	3	2	2	57.53	1 violation: MLOGP > 4.15	Low	– 3.26	0.85

MW Molecule Weight, Hacc Hydrogen Bond Acceptors, Hdon Hydrogen Bond Donors, Rbon Rotatable Bonds, TPSA Topological Polar, GIA Gastrointestinal Absorption, BS Bioavailability Score

Table 3 Toxicity of components

		‡bSS	BA
Mol ID	–	222,284	64,971
LD50 (mg/kg)	–	890	2610
Toxicity			
Classification	Target	Prediction	Prediction
Organ toxicity	Hepatotoxicity	Inactive	Inactive
Organ toxicity	Neurotoxicity	Inactive	Inactive
Organ toxicity	Nephrotoxicity	Inactive	Inactive
Organ toxicity	Respiratory toxicity	Active	Active
Organ toxicity	Cardiotoxicity	Active	Active
Toxicity end points	Carcinogenicity	Active	Active
Toxicity end points	Immunotoxicity	Active	Active
Toxicity end points	Mutagenicity	Inactive	Inactive
Toxicity end points	Cytotoxicity	Inactive	Inactive
Toxicity end points	Ecotoxicity	Inactive	Inactive

edges, demonstrate the interactions between the ingredients and their corresponding target genes. The findings indicate that each target within the drug-target-disease interaction network interconnect with an average of 2.105 active ingredients of BS.

PPI

In this investigation, we carried out an extensive examination of 227 shared targets between BS and ES/CI. The STITCH 5.0 database was used for the PPI network construction (Fig. 2a). Afterward, we filtered out irrelevant nodes from this network to arrive at a refined PPI network. To further enhance our understanding of the interactions within this structure, we conducted an analysis using the STITCH database. In this visualization, targets were represented as nodes, while the lines linking them denote the

interactions occurring between pairs of targets. A variety of colors used in Fig. 2b reflect the various types of interactions identified within the network. Our analysis specifically targeted the betweenness centrality, degree, and closeness centrality of each node. This process resulted in the detection of 10 top target genes with notably high centrality values. These key genes included Fos Proto-Oncogene (*FOS*), Jun proto-oncogene (*JUN*), mitogen-activated protein kinase 1 (*MAPK1* or *JNK2*), heat shock protein 90 alpha family class A member 1 (*HSP90 AAI*), phosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit alpha (*PIK3 CA*), mitogen-activated protein kinase 8 (*MAPK8* or *JNK1*), mechanistic target of rapamycin (*mTOR*), peroxisome proliferator-activated receptor alpha (*PPARA*), ribosomal protein S6 kinase beta-1 (*RPS6 KB1*), and estrogen receptor 1 (*ESR1*) (Fig. 2c and Table 4).

Correlation analysis of key genes

To elucidate the relationships between ten prominent genes (*FOS*, *JUN*, *MAPK1*, *HSP90 AAI*, *PIK3 CA*, *MAPK8*, *MTOR*, *PPARA*, *RPS6 KB1*, and *ESR1*) and major genes associated with CI, specifically *APOE* and *MTHFR*, a gene correlation analysis was conducted using the GEPIA database. The analysis identified nine significant correlations between *APOE* and the selected top genes, including *FOS*, *JUN*, *HSP90 AAI*, *PIK3 CA*, *MAPK8*, *MTOR*, *PPARA*, *RPS6 KB1*, and *ESR1*. These correlations suggest a substantial interaction between *APOE* and these genes, indicating their potential involvement in the molecular pathways that underlie CI (Fig. 3). The significant associations observed may reflect the biological mechanisms through which *APOE* influences cognitive function and the risk of cognitive decline. Moreover, *MTHFR* exhibited significant correlations with all ten of the top genes analysed, particularly in the hippocampus, a critical region for memory and learning.

Fig. 2 Protein–Protein Interaction (PPI) Network of BS and ES/CI Targets. **a** Initial PPI network of 227 shared targets between BS and ES/CI, generated using STITCH 5.0 database (Homo sapiens filter). **b** Refined PPI network after removing irrelevant nodes, with nodes as targets and edges colored by interaction type (e.g., binding, activation), highlighting connectivity. **c** Top 10 key genes (e.g., FOS, JUN, MAPK1) ranked by centrality measures (degree, betweenness, closeness) using CytoNCA in Cytoscape, indicating their critical roles in ES/CI pathology

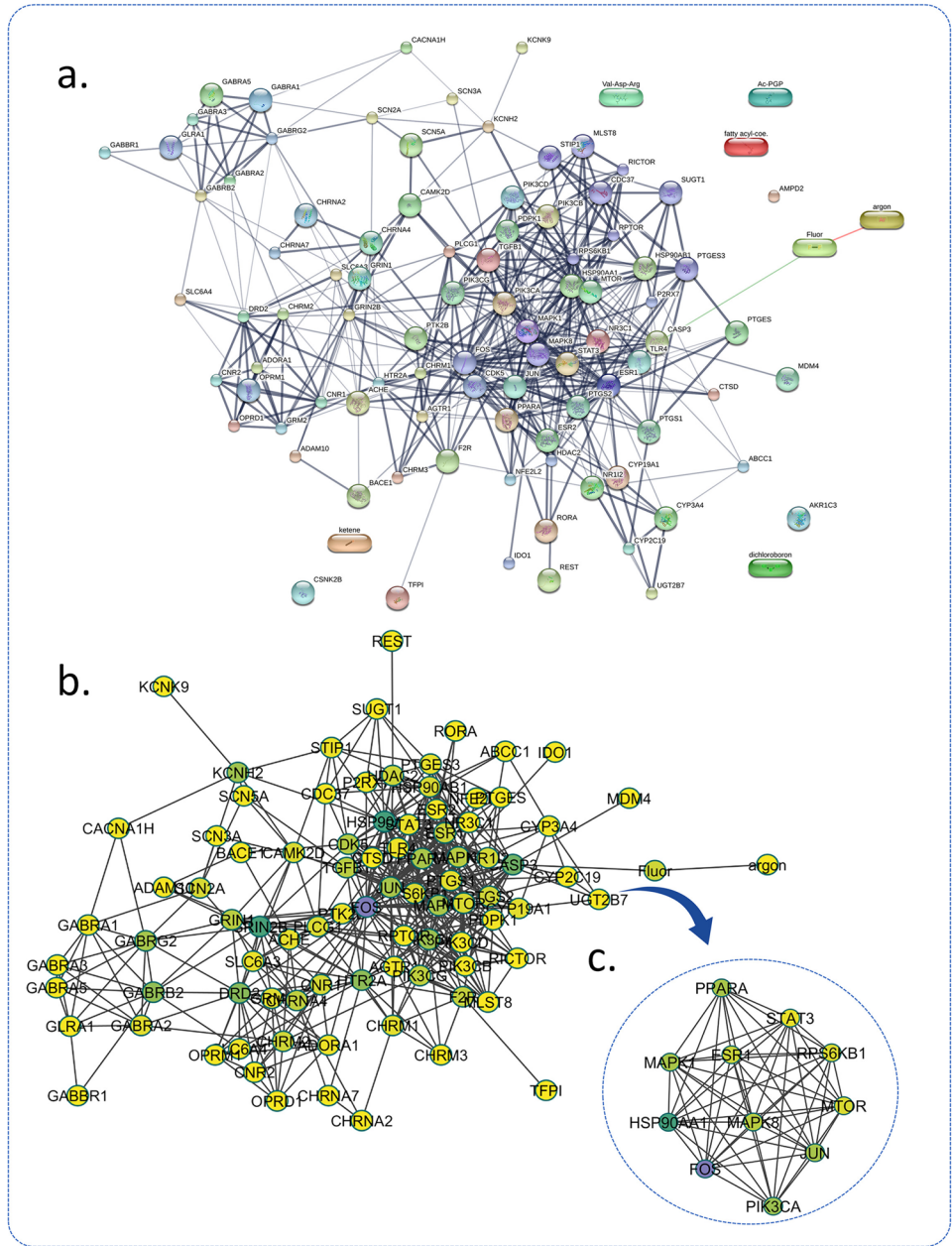


Table 4 Top ten key targets of protein–protein interaction

No	Genes	Degree	Betweenness	Closeness
1	FOS	38.0	1743.9434	0.60273975
2	JUN	31.0	333.01566	0.52071005
3	MAPK1	29.0	320.60593	0.5116279
4	HSP90 AA1	28.0	726.78723	0.5146199
5	PIK3 CA	25.0	380.9825	0.5146199
6	MAPK8	24.0	260.72833	0.4888889
7	MTOR	23.0	127.68121	0.4888889
8	PPARA	22.0	322.0125	0.4888889
9	RPS6 KB1	22.0	120.50612	0.48087433
10	ESR1	22.0	191.7686	0.47826087

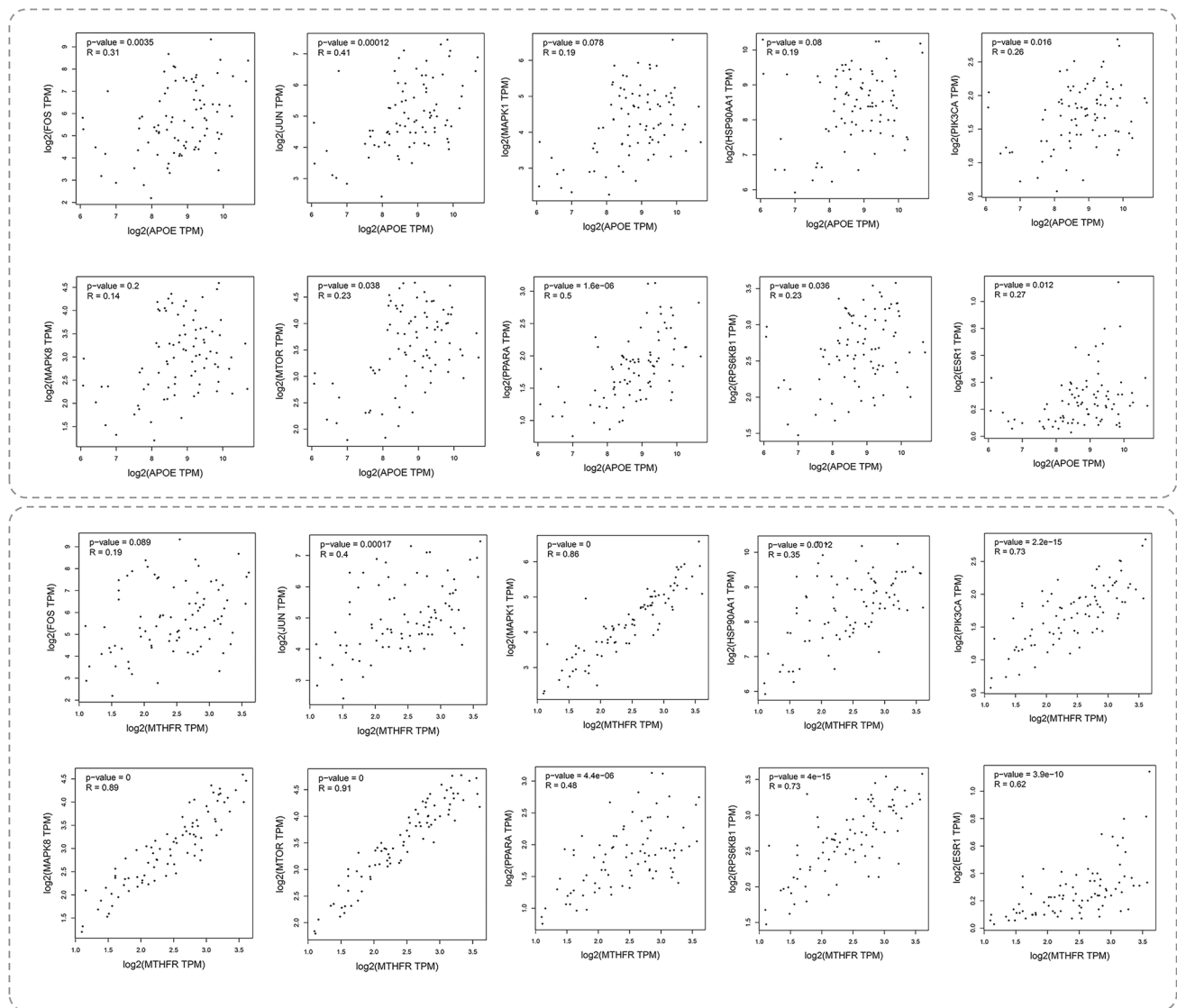


Fig. 3 Correlation Analysis of Key Genes with APOE and MTHFR in CI. Bar graphs depict Pearson correlation coefficients from GEPIA database analysis, showing significant associations between ten key genes (FOS, JUN, MAPK1, HSP90 AA1, PIK3 CA, MAPK8,

MTOR, PPARA, RPS6 KB1, ESR1) and major CI risk genes APOE and MTHFR in hippocampal tissue. APOE correlates with nine genes (all except MAPK1), while MTHFR correlates with all ten, underscoring their relevance to cognitive impairment mechanisms

This comprehensive correlation underscores the pivotal role of *MTHFR* in modulating the effects of these key genes on cognitive processes. Given *MTHFR*'s involvement in folate metabolism and its implications for neural function, these findings highlight its relevance in the context of CI (Fig. 3).

GO and KEGG

We conducted GO and KEGG pathway enrichment on the potential target genes to elucidate the underlying mechanisms by which the ingredients of BS may be effective in treating ES/CI. The results of these analyses, along with the enriched KEGG pathways, are illustrated in Fig. 4a–c

and Fig. 5, respectively. The overlapping targets are mainly distributed in cell components such as neuron projections, nuclear lumen, nucleoplasm, membrane bounded cell projections, cell projections, etc. They are involved in molecular functions such as response to oxygen containing compound, cell death, and response to external stimulus, response to organic substance, and positive regulation of nitrogen compound metabolic process. The results indicated that the mechanism of BS against PE induced cognitive disorders mainly involves enzyme binding, identical protein binding, adenyly nucleotide binding, adenyly ribonucleotide binding, ATP binding, etc. Notably, the KEGG analysis exposed pathways like Colorectal cancer, Prolactin signaling pathway,

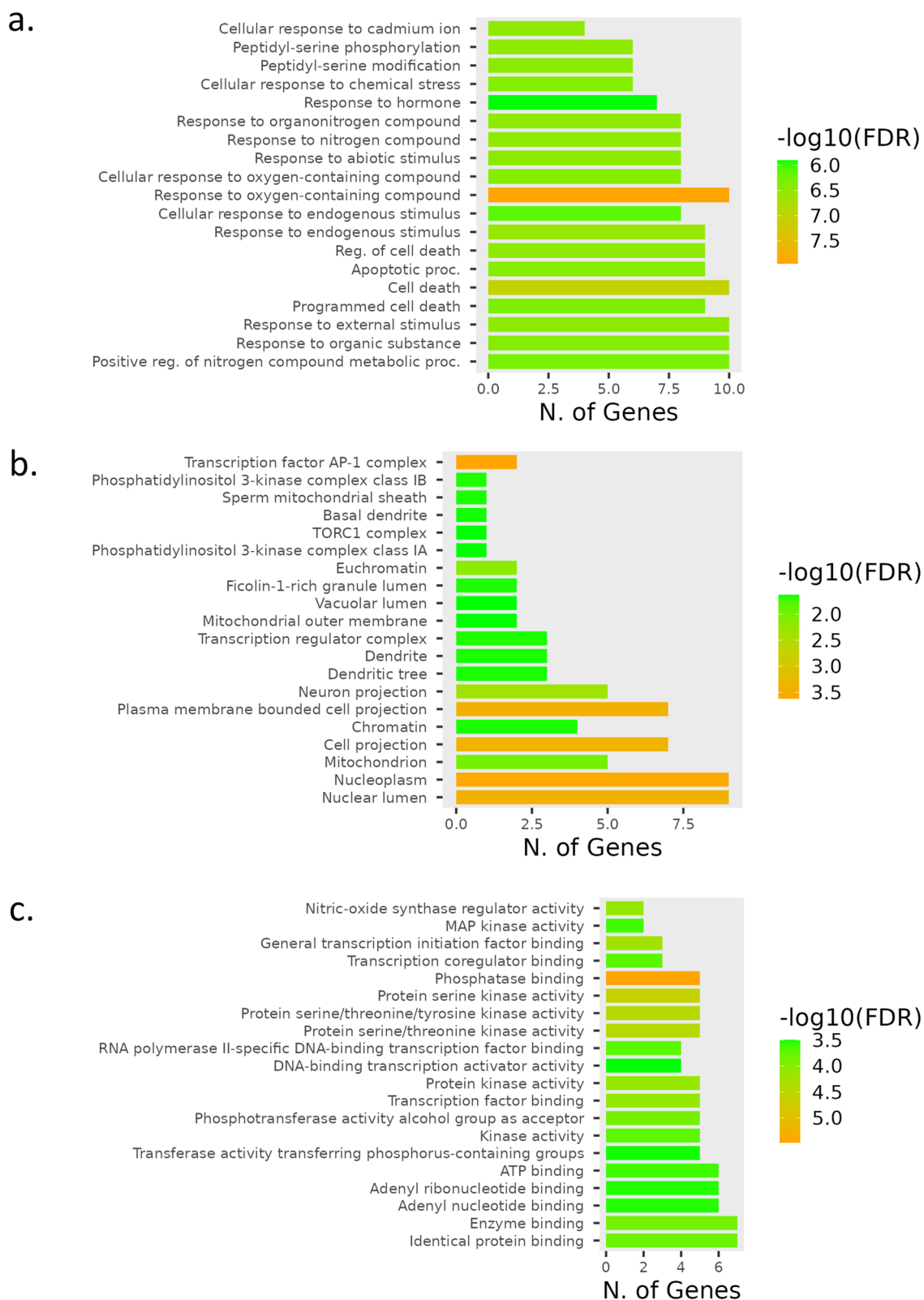
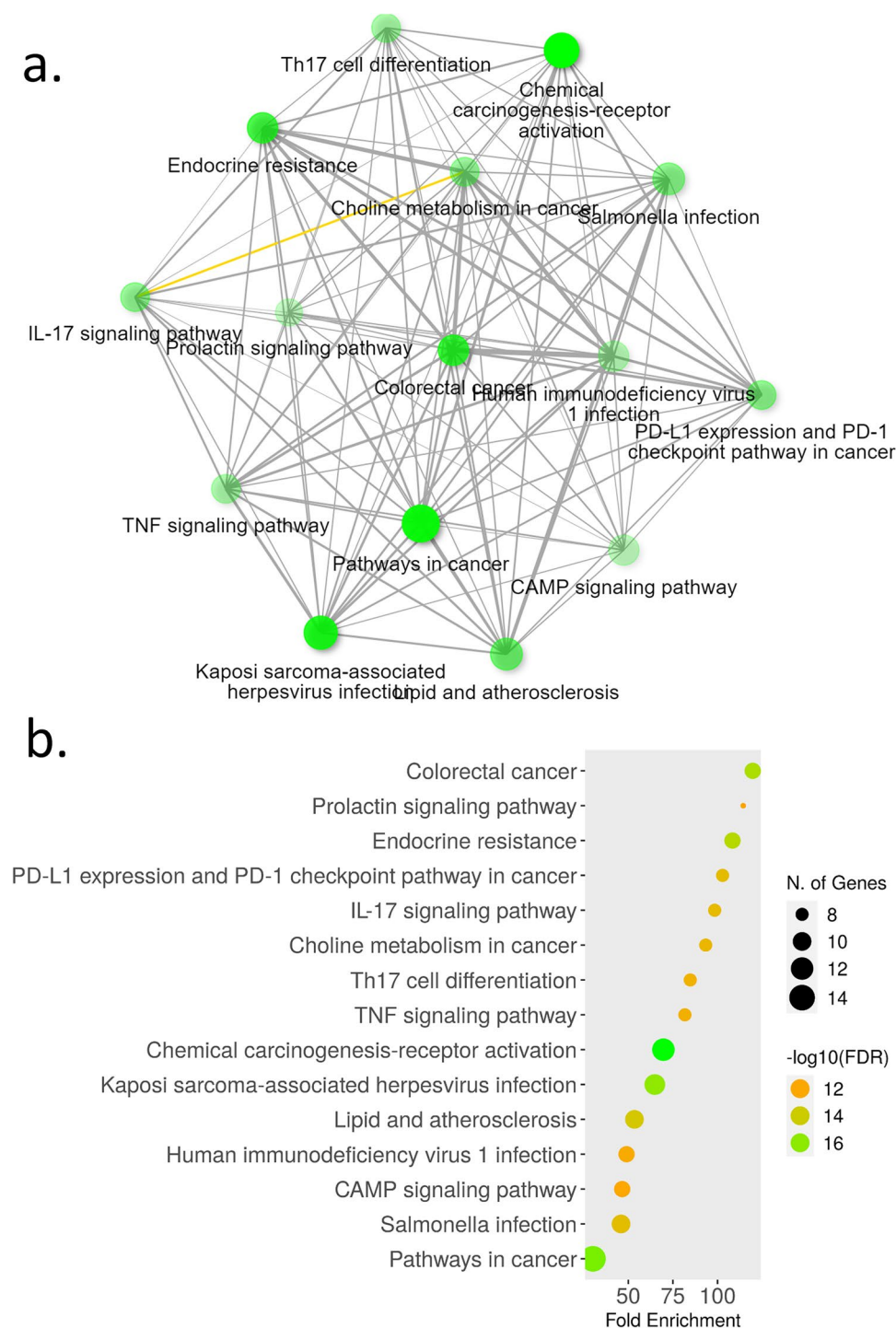


Fig. 4 Gene Ontology (GO) Analysis of BS Targets in ES/CI Treatment. Bar plots display the top 10 enriched GO terms ($p < 0.05$) for 277 overlapping BS and ES/CI targets using ShinyGO 0.77. **a** Biological processes, including response to oxygen-containing compounds

and cell death. **b** Cellular components, such as neuron projections and nucleoplasm. **c** Molecular functions, including enzyme binding and ATP binding, illustrating BS's functional impact on ES/CI

Fig. 5 Kyoto Encyclopedia of Genes and Genomes (KEGG) Enrichment Analysis of BS Targets in ES/CI. Dot plot shows significantly enriched KEGG pathways ($p < 0.05$, $FDR < 0.05$) for 277 shared targets, including IL-17 signaling, TNF signaling, and pathways in cancer, using ShinyGO 0.77. Dot size reflects gene count, and color indicates significance, highlighting inflammatory and survival pathways modulated by BS



Endocrine resistance, PD-L1 expression and PD-1 checkpoint pathway in cancer, IL-17 signaling pathway, Choline metabolism in cancer, Th17 cell differentiation, TNF signaling pathway, Chemical carcinogenesis-receptor activation, Kaposi sarcoma-associated herpesvirus infection, Lipid and atherosclerosis, Human immunodeficiency virus 1 infection, CAMP signaling pathway, Salmonella infection, Pathways in cancer, respectively (Fig. 5).

Signalling pathways

Among the various pathways suggested, two key molecular pathways have been recognized as crucial for the treatment of ES/CI via bioactive components: the IL-17 signalling pathway and the TNF signalling pathway (see Fig. S1 a & b). The gene targets that may be relevant for treating ES/CI with the bioactive components of BS are marked in red. Further

investigation indicated that the molecular mechanisms predominantly involve immune and inflammatory pathways, with a specific emphasis on the classical IL-17 and TNF signaling pathways. The critical targets within these pathways include Fos, Jun, caspase-3, PI3 K, AKT, and COX2.

Molecular docking

We employed molecular docking analysis to determine the interactions between five key genes of the therapeutic compound and two ingredients. This technique allowed us to identify the most steady binding configurations between the receptors and ligands, as confirmed by more negative Vina scores (VS) (Yang Liu et al. 2020). The docking results for bSS and BA with various receptors indicate distinct binding affinities that may influence their therapeutic potential. For bSS, the strongest interaction was observed with HSP90 AA1 (VS = − 9.2 kcal/mol). PIK3 CA also showed a strong binding affinity (VS = − 8.8 kcal/mol), while MAPK1 demonstrated a notable interaction (VS = − 8.5 kcal/mol). In contrast, the interactions with FOS and JUN were weaker, with binding energies of VS = − 6.2 kcal/mol and VS = − 5.8 kcal/mol, respectively. In comparison, the docking results for BA reveal a different profile. PIK3 CA displayed the highest binding affinity with a score of (VS = − 9.7 kcal/mol), indicating an even stronger interaction than that observed for BS. MAPK1 followed with a binding energy of (VS = − 8.2 kcal/mol), while HSP90 AA1 showed a moderate affinity at (VS = − 8.0 kcal/mol). The interactions with FOS and JUN were again weaker, with scores of (VS = − 5.4 kcal/mol) and (VS = − 5.3 kcal/mol), respectively (Fig. 6, Table 5).

Findings from in vivo study

Behavioral tests

The OFT revealed significant differences between study groups in the number of area crossings ($p < 0.05$, Fig. 7a) and latency (time taken) in the central region ($p < 0.05$, Fig. 7b). According to Fig. 7, compared to the CON mice, the PTZ-treated animals exhibited a decrease in the number of times they crossed different areas and an increase in the latency to enter the central region of the arena. Treatment with doses of 200 and 400 mg/kg of BS (the tested compound) and 100 mg/kg of VLP (the reference compound) were able to improve these measures in the OFT, suggesting a potential beneficial effect on the animals' exploratory behavior and anxiety-like behavior.

The results of the MWM task in terms of escape latency (the time taken by the animals to find the hidden platform) showed no significant differences were seen between the groups on days 1 and 2 of the task ($p > 0.05$). However,

from days 3 to 4 of the task, the mice in the PTZ-induced group exhibited enhanced escape latency compared to the control (CON) group ($p < 0.05$, Fig. 7c), indicating impaired spatial learning. In contrast, the mice in the PTZ + BS2, PTZ + BS3, and PTZ + VLP groups had shorter escape latencies compared to the PTZ-induced group that received the vehicle ($p < 0.05$, Fig. 7c), suggesting that treatment with the BS at doses of 200 and 400 mg/kg, as well as VLP at 100 mg/kg, were able to ameliorate the spatial learning deficits induced by PTZ exposure. In the probe trial, which assesses spatial memory, the passing times in the target quadrant was reduced in the PTZ-induced animals in comparison with the CON group ($p < 0.05$). Nevertheless, a notable raise in passing times in the target quadrant was observed in the PTZ + BS2, PTZ + BS3, and PTZ + VLP groups compared to the PTZ-induced group that received the vehicle ($p < 0.05$, Fig. 7d). The MWM results indicate that exposure to PTZ induced deficits in spatial learning and memory. Importantly, the administration of the BS at doses of 200 and 400 mg/kg and VLP at 100 mg/kg were effective treatments.

qPCR

We evaluated the effects of BS on the gene expression of hippocampal *Fos*, *Jun*, *Casp3*, *PI3 K*, *AKT*, and *COX2*. Our results showed the upregulation of *Fos*, *Jun*, *Casp3*, and *COX-2* and downregulation of *AKT* and *PI3 K* in the hippocampus of animals exposed to PTZ ($p < 0.05$, Fig. 7a–f). Importantly, administration of BS at doses of 200 and 400 mg/kg or VLP at 100 mg/kg could improve the gene expression of hippocampal *Fos*, *Jun*, *Casp3*, *PI3 K*, *AKT*, and *COX2* in the PTZ-treated group ($p < 0.05$, Fig. 8a–f). These findings suggest that BS and VLP treatments were able to regulate the expression of components of IL-17 and TNF signaling pathways.

ELISA

We evaluated the impacts of BS on the protein concentrations of hippocampal $\text{TNF}\alpha$, IL6, and IL1 β . The results of the ELISA presented a raise in the protein concentrations of $\text{TNF}\alpha$, IL6, and IL1 β in the hippocampus ($p < 0.05$, Fig. 9a–c). Notably, administration of BS at doses of 200 and 400 mg/kg or VLP at 100 mg/kg could improve the protein levels of hippocampal $\text{TNF}\alpha$, IL6, and IL1 β in the PTZ-treated groups ($p < 0.05$, Fig. 7a–f). These findings suggest that BS and VLP treatments were able to modulate the downstream proteins involved in the IL-17 and TNF signaling pathways, which may have contributed to the observed neuroprotective effects in the PTZ-induced model.

Table 5 Molecular docking

L	R	VS	CV (Å ³)	C (x, y, z)	Contact residues
bSS	FOS	− 6.2	24	− 3, 14, − 14	Chain E: ILE183 LEU186 LEU187 LYS188 GLU189 LYS190 GLU191 LYS192 LEU193 GLU194 PHE195 ILE196 LEU197 ALA198
	JUN	− 5.8	15	38, 4, − 16	Chain F: SER278 ARG279 LYS280 ARG281 LYS282 LEU283 GLU284 ARG285 ILE286 ALA287 ARG288 LEU289 GLU290 GLU291 LYS292 VAL293 LYS297
	MAPK1	− 8.5	603	67, 12, 7	Chain A: TYR30 ILE31 GLY32 GLU33 GLY34 ALA35 TYR36 VAL39 SER41 ARG50 ALA52 ILE53 LYS54 GLU71 ILE84 ILE103 GLN105 ASP106 LEU107 MET108 GLU109 THR110 ASP111 TYR113 LYS114 LYS117 LYS151 SER153 ASN154 LEU156 CYS166 ASP167 PHE168
	HSP90 AA1	− 9.2	2250	7, 11, 24	Chain A: PHE22 LEU48 ASN51 SER52 ASP54 ALA55 ASP57 LYS58 ILE91 ASP93 ILE96 GLY97 MET98 ASP102 LEU103 ASN106 LEU107 GLY108 ILE110 ALA111 GLY135 VAL136 GLY137 PHE138 TYR139 VAL150 TRP162 THR184 VAL186
	PIK3 CA	− 8.8	3533	73, 56, 79	Chain A: ASP138 ASN142 ARG398 ALA399 TYR432 THR433 ASP434 ASP608 PRO609 MET610 ARG612 GLY613 VAL616 TYR641 GLU642 GLN643 TYR644 ASP646 ASN647 LEU648 LEU649 ARG651
BA	FOS	− 5.4	6	18, 10, − 22	Chain E: THR169 GLN171 LEU172 GLU173 ASP174 GLU175 LYS176 SER177 ALA178 LEU179 GLN180 ILE183
	JUN	− 5.3	15	38, 4, − 16	Chain F: SER278 ARG281 LYS282 LEU283 GLU284 ARG285 ILE286 ARG288 LEU289 GLU290 VAL293
	MAPK1	− 8.2	603	67, 12, 7	Chain A: TYR30 ILE31 GLY32 GLU33 GLY34 ALA35 TYR36 VAL39 ALA52 LYS54 ILE84 GLN105 ASP106 LEU107 MET108 GLU109 THR110 ASP111 LEU112 TYR113 LYS114 LEU115 LYS117 LYS151 SER153 ASN154 LEU156 LYS164 ILE165 CYS166 ASP167
	HSP90 AA1	− 8.0	2250	7, 11, 24	Chain A: LEU29 GLU47 LEU48 ASN51 SER52 ASP54 ALA55 LEU56 ASP57 LYS58 ASP93 THR94 GLY95 ILE96 GLY97 MET98 ASP102 ASN106 LEU107 ILE110 ALA111 SER113 GLY132 GLN133 PHE134 GLY135 VAL136 GLY137 PHE138 TYR139 VAL150 THR152 HIS154 GLY183 THR184
	PIK3 CA	− 9.7	7709	56, 75, 101	Chain A: SER-27 TYR-26 TYR-25 HIS-23 HIS-22 HIS-21 ASN170 GLU172 ILE181 GLU259 LYS271 TYR272 ARG274 SER275 MET278 LEU279 THR624 ASP625 ASP626 LYS627 LEU628 SER629 GLN630 ARG662 PHE666 LEU755 ASN756 PRO757 ALA758 HIS759 ARG818 GLU821 ASN822 ILE823 TRP824 GLN825 ASN826 ARG832 MET833 PRO835 TYR836 GLY837 CYS838 GLU849 GLN928

bSS β -sitosterol, BA Betulinic acid, L Ligand, R Receptors, VS Vina score, CV Cavity volume, C Center

Discussion

In this study, we investigated the molecular mechanisms underlying the therapeutic effects of BS for ES/CI comorbidity. Through an in silico approach, we identified a comprehensive set of 6253 genes associated with epilepsy, cognitive disorders, and inflammation. From this extensive dataset, we narrowed our focus to 277 genes that overlapped between the predicted targets of BS ingredients and disease-related genes. PPI analysis further refined our selection, highlighting ten key genes based on degree, betweenness, and closeness centralities, which were subsequently examined using GO and KEGG pathway enrichment analyses. To strengthen our findings, we assessed correlations between these key genes and established risk factors, such as *APOE* and *MTHFR*, and performed molecular docking studies. Finally, we validated the therapeutic potential of BS in an in vivo model of PTZ-induced ES/CI. Collectively, our results highlight BS as a promising candidate for treating this complex condition.

Our in vivo study confirmed the presence of ES/CI in PTZ-induced mice through behavioral assessments using the OFT and the MWM. These tests were selected due to their sensitivity in detecting ES/CI-related deficits. The OFT measures anxiety-like behavior, locomotor activity, and motor coordination, all of which are impaired by PTZ-induced seizures, whereas the MWM assesses spatial learning and memory, which are compromised in cognitive impairment. In PTZ-treated mice, the OFT revealed reduced exploration time and increased latency to enter the central zone, while the MWM demonstrated prolonged escape latencies and fewer target quadrant crossings, confirming the presence of ES/CI. Comorbidities associated with epilepsy are highly prevalent and often severe. Common comorbid conditions include mental health disorders such as depression and anxiety, as well as physical health issues like sleep disturbances and migraines. Moreover, individuals with epilepsy frequently experience cognitive decline, which manifests as difficulties in attention, memory, and information processing (Holmes 2015). CIs

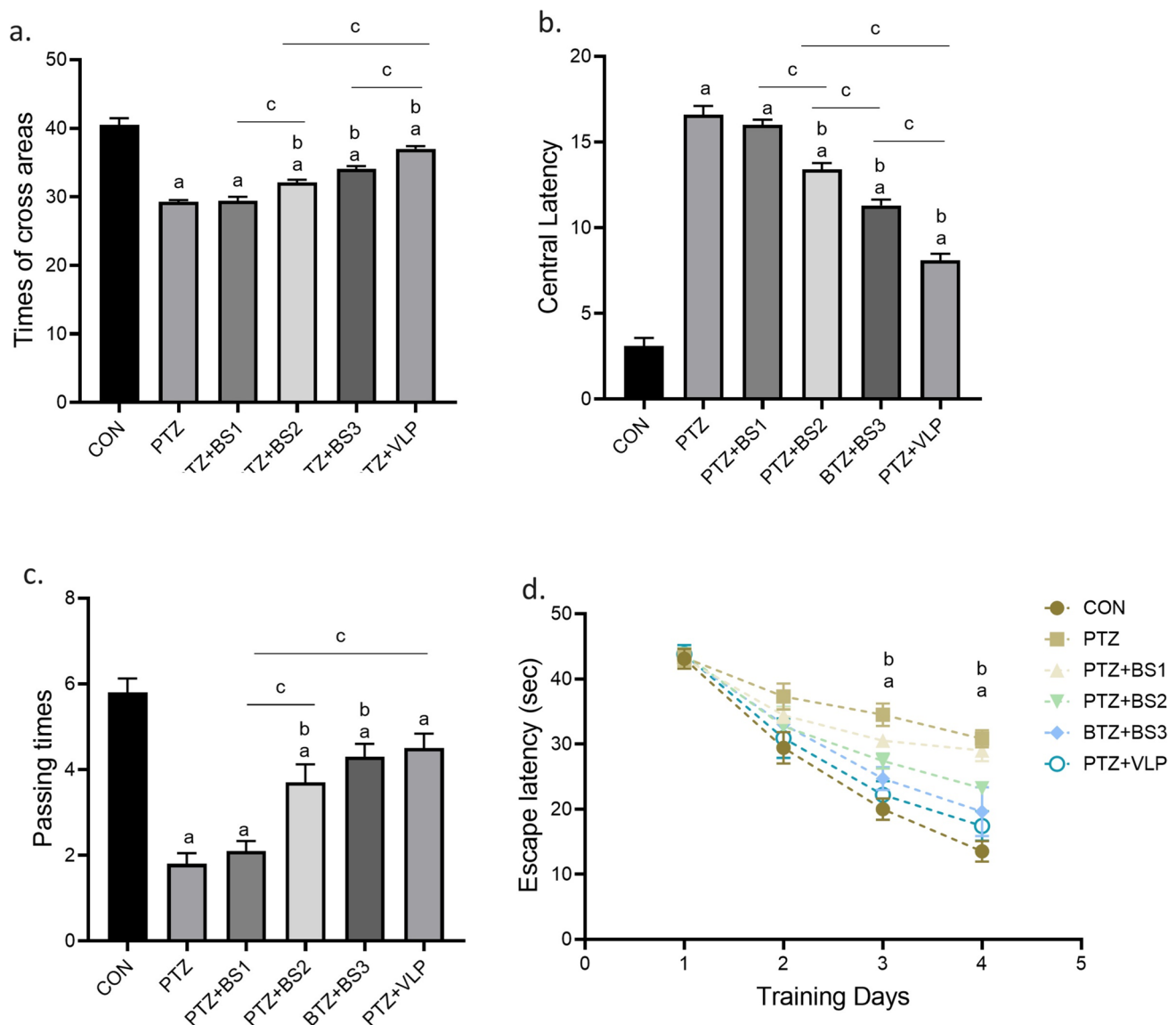


Fig. 7 Effects of BS on exploratory behavior and spatial learning/memory in PTZ-induced mice. Bar and line graphs show Open Field Test (OFT) and Morris Water Maze (MWM) results across six groups ($n = 12$): CON, PTZ, PTZ + VLP (100 mg/kg), PTZ + BS (100, 200, 400 mg/kg). **a** OFT area crossings, reduced in PTZ vs. CON, improved by BS (200, 400 mg/kg) and VLP. **b** OFT central latency, increased in PTZ, reduced by treatments. **c** MWM escape latency

over 4 days, prolonged in PTZ, shortened by BS (200, 400 mg/kg) and VLP. **d** MWM probe trial passing times, decreased in PTZ, increased by treatments. ^a Compared with CON group: $P < 0.05$, ^b Compared with PTZ group, and ^c Comparison between BS (three doses) and VLP treatment group. One-way ANOVA followed by post hoc Tukey

rank among the most prevalent and challenging comorbidities linked to epilepsy. These abnormalities can significantly affect an individual's cognitive functioning and overall quality of life, making them a critical concern in the management of epilepsy (Berg et al. 2008). Cognitive impairments rank among the most prevalent and challenging comorbidities linked to epilepsy. These abnormalities can significantly affect an individual's cognitive functioning and overall quality of life, making them a critical concern in the management of epilepsy (Malmgren

et al. 2008; Hsiang-Yu Yu et al. 2009). The hippocampus, a region of the brain particularly susceptible to damage, is often adversely affected by excitotoxic agents such as PTZ, a noncompetitive antagonist of gamma-Aminobutyric acid type A receptors ($GABA_A$ R), commonly utilized to induce seizure activity in animal models (Walker). This characteristic renders PTZ a suitable agent for modeling ES-induced CI. Understanding the molecular mechanisms of these conditions is essential for identifying potential therapeutics.



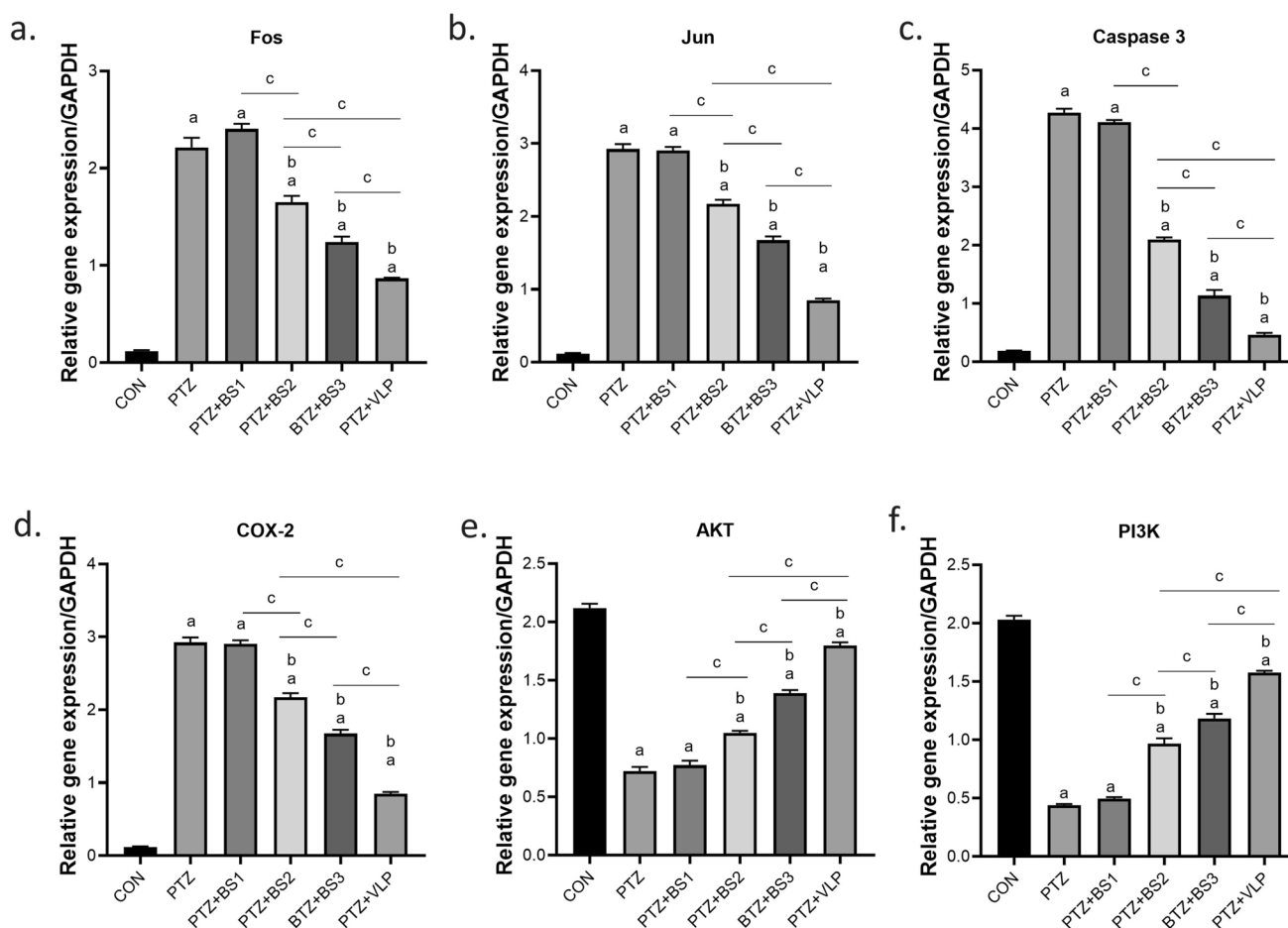


Fig. 8 Gene Expression Effects of BS in PTZ-Induced Mice Hippocampus. Bar graphs show qRT-PCR results ($n = 6/\text{group}$) for (a) Fos, (b) Jun, (c) Casp3, (d) PI3 K, (e) AKT, and (f) COX2 in CON, PTZ, PTZ + VLP (100 mg/kg), and PTZ + BS (100, 200, 400 mg/kg) groups. PTZ upregulates Fos, Jun, Casp3, COX2 and down-

regulates PI3 K, AKT; BS (200, 400 mg/kg) and VLP reverse these changes. Data normalized to GAPDH, analyzed by one-way ANOVA with Tukey's test ($p < 0.05$). ^a Compared with CON group: $P < 0.05$, ^b Compared with PTZ group, and ^c Comparison between BS (three doses) and VLP treatment group using

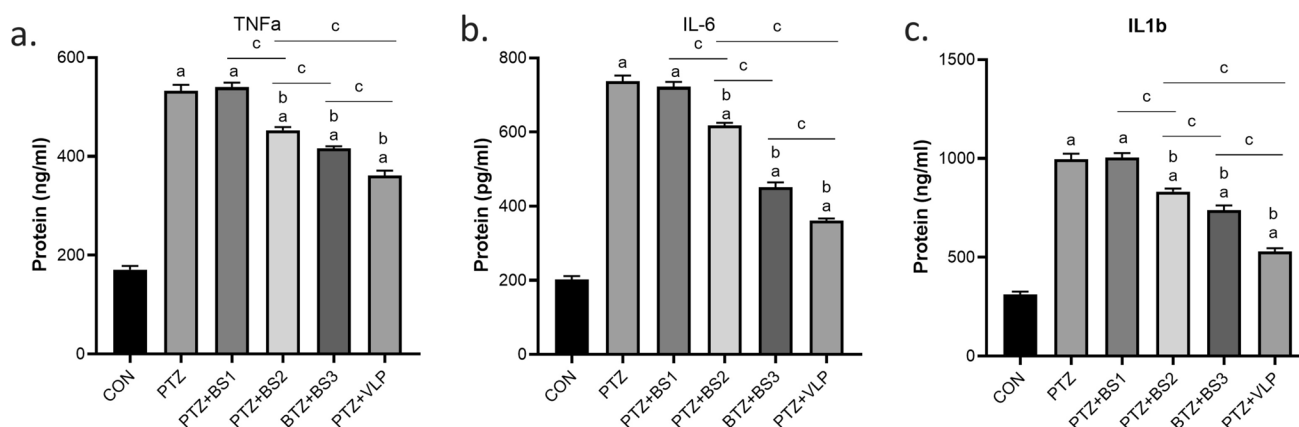


Fig. 9 Protein Level Effects of BS in PTZ-Induced Mice Hippocampus. Bar graphs display ELISA results ($n = 6/\text{group}$) for (a) TNF- α , (b) IL-6, and (c) IL-1 β in CON, PTZ, PTZ + VLP (100 mg/kg), and PTZ + BS (100, 200, 400 mg/kg) groups. PTZ increases all

cytokines; BS (200, 400 mg/kg) and VLP reduce levels. Statistical significance ($p < 0.05$) via one-way ANOVA with Tukey's test: ^a vs. CON, ^b vs. PTZ, ^c between treatments, indicating BS's anti-inflammatory effects

Recent advancements in bioinformatics have enabled the discovery of potential therapeutic compounds by analyzing gene expression and molecular pathways, facilitating the prediction and identification of promising drug candidates (Mokhtari and Ayman 2024). In this study, we employed computational analysis to predict the genes and pathways involved in ES-induced cognitive impairment. Our findings suggest a strong link between ES-induced brain damage and cognitive impairment through inflammatory pathways, as we identified a significant number of overlapping genes among these three conditions. Moreover, our analyses revealed a core group of 10 out of 277 target genes associated with ES/CI that can be modulated by the bioactive ingredients present in BS. These key target genes include *FOS*, *JUN*, *MAPK1*, *HSP90 AA1*, *PIK3 CA*, *MAPK8*, *MTOR*, *PPARA*, *RPS6 KBI*, and *ESR1*. GO analysis indicated that the proteins encoded by these genes are distributed across various cellular components, including neuron projections, the nuclear lumen, nucleoplasm, membrane-bounded cell projections, and cell projections. Functionally, these proteins participate in key biological processes such as responses to oxygen-containing compounds, cell death, responses to external stimuli, responses to organic substances, and the positive regulation of nitrogen compound metabolic processes. These findings suggest that the pathophysiology of ES/CI primarily involves molecular interactions such as enzyme binding, identical protein binding, ATP binding, adenylyl ribonucleotide binding, and adenylyl nucleotide binding. Furthermore, the identified genes are associated with key neuroinflammatory pathways, including the IL-17 and TNF signaling pathways, both of which play crucial roles in neuronal dysfunction. The critical molecular targets within these pathways—*Fos*, *Jun*, *caspase-3*, *PI3 K*, *AKT*, and *COX2*—underscore their potential as therapeutic targets for ES/CI.

To further validate the role of these genes in the CNS and their involvement in the pathophysiology of ES and CI, we examined their molecular functions in more detail. The *FOS* family, which includes *Fos*, *Fos-B*, *Fra2* (*FOSL2*), and *Fra1* (*FOSL1*), encodes proteins with a leucine zipper motif that enables dimerization with *JUN* family members, forming the activator protein 1 (AP-1) complex—a transcription factor critical for cellular response mechanisms (Lorençatto et al. 2006). In the CNS, *Fos* expression is closely linked to neuronal depolarization and is influenced by glial cells such as astrocytes, microglia, and oligodendrocytes, which contribute to inflammation, repair, plasticity, and brain microenvironment homeostasis (Lorençatto et al. 2006; Balázs 2010). Notably, *Fos* expression is implicated in cell division and differentiation following brain injury (Lara Aparicio et al. 2022).

The *PIK3 CA* gene encodes the catalytic subunit p110 α of class I PI3 K (Furet et al. 2013; Jean J. Zhao et al. 2005). PI3

K activation, triggered by phosphorylated tyrosine kinase receptors, initiates a phosphorylation cascade leading to the activation of *AKT*, which in turn phosphorylates *mTOR*. The downstream effects of *mTOR* regulate genes such as *Rps6kb1* and *ribosomal protein S6* (*Rps6*). Inhibition of PI3 K counteracts *mTOR* activation, suggesting its role in neuroprotection (Wei et al. 2018). The PI3 K/Akt/*mTOR* signaling pathway plays a vital role in neuronal growth, proliferation, survival, and inflammation, and its dysregulation has been associated with ES-induced neurodegeneration (Peng et al. 2022; Hodges and Lugo 2020). The MAPK family, encompassing the p38, extracellular signal-regulated kinase (ERK), and c-Jun N-terminal kinase (JNK) pathways (Nateri et al. 2007), serves as another crucial regulatory system in neuronal excitability. Both *mTOR* and MAPK are essential regulators of synaptic excitability, and their dysregulation has been implicated in ES/CI in both animal models and human disorders (Krab et al. 2008; Kumar et al. 2005). The localization of mRNAs encoding these biomarkers near synapses suggests that their dysregulation may contribute to synaptic dysfunction and impaired neuronal signaling (Pernice et al. 2016). Heat shock proteins (HSPs), also known as stress proteins, play a pivotal role in cellular responses to stressors such as extreme temperatures, nutrient deprivation, and hypoxia (Bond and Schlesinger 1987). Among them, *HSP90* is a key molecular chaperone involved in protein folding and stability (Ou et al. 2014). It is one of the most abundantly expressed proteins in cells, maintaining proteostasis under stress conditions (Whitesell and Lindquist 2005). Mammals express two primary subtypes: *HSP90 β* (*HSP90 AB1*), constitutively active, and *HSP90 α* (*HSP90 AA1*), an inducible form found predominantly in the brain (Ou et al. 2014; Bohush et al. 2019). The *HSP90 AB1* gene is particularly relevant in the immune microenvironment of neurodegenerative disorders (Lai et al. 2022). Notably, *HSP90 AB1* has been implicated in astrocyte-mediated autophagic degradation of amyloid-beta ($A\beta$) and tau via chaperone-mediated autophagy (Gonzalez-Rodriguez et al. 2021). *HSP90* also plays a role in regulating tau phosphorylation, a process associated with neurodegenerative diseases (Bohush et al. 2019). Elevated *HSP90* expression has been detected in neurons and glial cells in the dentate gyrus and CA1 regions in temporal lobe epilepsy (Kandratavicius et al. 2014), and inhibitors of *HSP90* have demonstrated neuroprotective effects in animal models (Luo et al. 2010).

Additionally, *PPAR α* , a nuclear receptor belonging to the PPAR family, is involved in energy metabolism and has been shown to exhibit anticonvulsive properties through its agonist, fenofibrate (Porta et al. 2009). *PPAR α* regulates genes encoding enzymes critical for neurotransmitter metabolism and amino acid processing in the brain (Cullingford 2004). Furthermore, it modulates inflammatory responses and oxidative stress cascades, reinforcing

its neuroprotective role (Cabrero et al. 2002). Other key targets include *Casp3*, a cysteine protease associated with apoptosis and neuronal loss following status epilepticus (Henshall et al. 2000), and *ESRI*, an estrogen receptor with neuroprotective properties, particularly in synaptic plasticity and neuronal resilience, which has been implicated in epilepsy treatment (Wang et al. 2022). Taken together, these findings suggest that ES-induced cognitive impairment arises from a complex interplay of neuroinflammation, synaptic dysfunction, oxidative stress, and apoptotic signaling. Disruptions in these molecular processes contribute to neuronal damage and cognitive decline associated with ES. By elucidating these interconnected mechanisms, our study advances the broader understanding of neuroinflammation's role in ES/CI, aiding the development of novel therapeutics. Importantly, targeting these pathological pathways with anti-inflammatory herbs like BS demonstrated a promising therapeutic strategy in our study to mitigate neurodegeneration and enhance cognitive outcomes in ES, paving the way for future research into anti-inflammatory and neuroprotective interventions.

To validate the involvement of these genes in the pathogenesis CI, we assessed the correlation of expression of these genes with *APOE* and *MTHFR*, as key genes associated with CI, playing crucial roles in neurodegenerative processes (Barati et al. 2022). *APOE* is widely recognized for its involvement in lipid metabolism, neuroinflammation, and synaptic plasticity (Meifang Xu et al. 2025). *MTHFR*, on the other hand, regulates folate metabolism and DNA methylation, processes essential for neural function and cognitive health (Román et al. 2019; Roussotte et al. 2017). *APOE* showed strong associations with nine of these genes, suggesting its role in shared molecular pathways linked to CI, while *MTHFR* correlated with all ten, particularly in the hippocampus, emphasizing its importance in cognitive function. These findings highlight the potential mechanistic interplay between *APOE*, *MTHFR*, and these key genes in CI, offering insights into their relevance for therapeutic interventions. As BS can modulate these target genes, it may serve as a potential treatment for CI, warranting further investigation into its neuroprotective properties.

To validate our *in silico* predictions, we conducted an *in vivo* study using male C57BL/6 J mice exposed to PTZ-induced ES. We assessed hippocampal dysregulation of key genes (*FOS*, *JUN*, *CASP3*, *PI3 K*, *AKT*, *COX2*) and inflammatory cytokines (*TNFα*, *IL-6*, *IL-1β*) in ES/CI animals. BS treatment (100–400 mg/kg) significantly increased center time and distance traveled in OFT, indicating improved motor function and reduced anxiety, and shortened escape latencies in MWM, reflecting enhanced cognitive recovery. Consistent with computational analyses, qRT-PCR and ELISA revealed upregulated *FOS*, *JUN*, *CASP3*, and *COX2*, downregulated *PI3 K*/*AKT*, and elevated *TNFα*, *IL-6*, and

IL-1β protein levels in PTZ-treated mice, underscoring neuroinflammation's role.

Using only male mice ensured result consistency by minimizing hormonal variability known to influence seizure susceptibility and cognition (Velíšková and DeSantis 2013). Recurrent ES can cause irreversible brain damage, contributing to CI in 70–80% of epilepsy patients, marked by cognitive slowing, memory loss, and attention deficits (Hermann et al. 2010; Lodhi and Agrawal 2012). ES arise from aberrant activity of voltage- and transmitter-gated ion channels, causing neuronal hyperexcitability and subsequent structural/functional brain changes (Fisher et al. 2017; Novak et al. 2022). Frequent ES trigger oxidative stress, neuronal death (notably in the hippocampus and entorhinal cortex), reduced BDNF expression, impaired neurogenesis, and neuroinflammation (van Rijckevorsel 2006; Holmes 2015). Neuroinflammation significantly contributes to CI in epilepsy, initiated by neural network disruption and synaptic loss post-injury, driving inflammatory cascades (Ngadimon et al. 2024). Human studies link CI to inflammatory pathways, glycolipid alterations, and depression (Arend et al. 2018), while PTZ-treated animals exhibit memory deficits tied to dysregulated inflammatory markers (e.g. interferon- γ [*IFN- γ*], toll-like receptor-4 [*TLR-4*], *CCL2*, *TNFα*, *IL-1*) and neurotransmitter imbalances (acetylcholinesterase [*AChE*], glutamate, GABA) (Kundap et al. 2019). Our PTZ model confirmed elevated hippocampal inflammation and CI, reversible by BS, suggesting that targeting neuroinflammation could alleviate epilepsy-related CI.

In this study, we explored the molecular mechanisms through which BS targets ES/CI. We began by investigating the ingredients of BS using the TCMSP database, initially identifying 39 bioactive compounds. To refine our selection, we applied stringent pharmacokinetic criteria, including $OB \geq 30\%$, $DL \geq 0.18$, and BBB permeability ≥ -0.3 , ensuring the inclusion of compounds with potential CNS activity. Based on these criteria, bSS and BA emerged as promising candidates with favorable ADME properties, while other compounds were excluded due to poor bioavailability or their inability to cross the BBB. We subsequently examined the ADMET properties of each ingredient. Although both compounds violated Lipinski's Rule of Five—suggesting potential challenges in oral absorption and permeability—they demonstrated good GI absorption, as indicated by their GI scores. Toxicological assessments further supported their safety profiles, showing relatively low acute toxicity. This safety profile was corroborated *in vivo*, where BS administration (100, 200, and 400 mg/kg) to PTZ-treated mice over 21 days showed no overt adverse effects. General health indicators, body weight, feeding behavior, and grooming, remained stable compared to control and PTZ-only groups, with monitoring conducted daily to ensure no subtle toxicity emerged. These findings reinforce BS's potential as a safe,

CNS-active intervention, complementing its molecular targeting of ES/CI pathways.

To the best of our knowledge, this study is the first to elucidate the involvement of key molecular pathways in ES/CI pathogenesis in the context of BS treatment. TCM has been extensively investigated for its potential in managing epilepsy and its associated CI (Tao et al. 2023). TCMs exert their therapeutic effects through intricate interactions among multiple bioactive compounds and targets, presenting a complex challenge in comprehensively deciphering their mechanisms (Shao Li 2016; Xiang et al. 2024). Among these, BS has a rich history of medicinal use in Asia, known for its significant influence on the immune and nervous systems (He et al. 2024). It exhibits a broad range of therapeutic effects, including neuroprotection, antidepressant, sedative, analgesic, and anticonvulsant properties (Du et al. 2020). Recent studies highlight the neuroprotective potential of BS in neurodegenerative diseases, including Parkinson's disease (PD), through active ingredients such as bSS, BA, kaempferol, palbinone, paeoniflorin, and (+)-catechin, as identified in an *in silico* study by Du et al. (Du et al. 2020). bSS, a widely distributed phytosterol, has demonstrated strong antioxidant, anti-inflammatory, and anti-cancer effects within the CNS (Chao Yang et al. 2013; Sharma et al. 2021). Notably, it effectively penetrates brain tissue following systemic administration. For instance, in a mouse model of brain ischemia, bSS was detected in the cerebral cortex 24 h after intraperitoneal injection, confirming its ability to cross the BBB. Additionally, bSS integrates into neuronal cell membranes, potentially modulating brain cell function. Research has shown that bSS reduces cerebral infarction and brain edema volumes, decreases neuronal apoptosis, and improves neurological function (Xiuling Tang et al. 2024). Similarly, BA, a natural pentacyclic triterpenoid, exhibits a broad spectrum of biological functions, including antioxidant, anti-inflammatory, and anti-tumor activities, significantly affecting brain disorders. However, its therapeutic potential is limited by poor water solubility and low bioavailability, posing challenges in BBB penetration (Yong Li et al. 2022b). Despite these limitations, our findings confirm that BS, particularly its active ingredients, holds substantial promise as a therapeutic agent for CNS disorders.

Pathway analysis identified ten key genes—*FOS*, *JUN*, *MAPK1*, *HSP90 AA1*, *PIK3 CA*, *MAPK8*, *MTOR*, *PPARA*, *RPS6 KBI*, and *ESR1*—implicated in ES/CI pathogenesis and modulated by BS. KEGG analysis further highlighted the *IL-17* and *TNF* signaling pathways, with critical targets (*FOS*, *JUN*, *CASP3*, *PI3 K*, *AKT*, and *COX2*) associated with inflammation and cellular survival. Molecular docking studies demonstrated strong binding affinities (*scores* < − 5 kcal/mol) for BS components bSS and BA, indicating their potential for direct molecular interactions. bSS exhibited the highest affinities with *HSP90 AA1* (− 9.2 kcal/mol), *PIK3*

CA (− 8.8 kcal/mol), and *MAPK1* (− 8.5 kcal/mol), while its binding to *FOS* (− 6.2 kcal/mol) and *JUN* (− 5.8 kcal/mol) was weaker. Similarly, BA demonstrated the strongest interaction with *PIK3 CA* (− 9.7 kcal/mol), followed by *MAPK1* (− 8.2 kcal/mol) and *HSP90 AA1* (− 8.0 kcal/mol), with lower affinities for *FOS* (− 5.4 kcal/mol) and *JUN* (− 5.3 kcal/mol). To validate these *in silico* predictions, we conducted an *in vivo* study to assess the neuroprotective effects of BS on ES/CI and its molecular targets. Administration of BS (200 and 400 mg/kg) significantly improved exploratory locomotion, spatial learning, and memory in PTZ-treated male mice, as assessed by the OFT and MWM tests. These cognitive enhancements were comparable to those observed with VLP, 100 mg/kg, further supporting BS's potential to mitigate ES/CI-related impairments. Furthermore, the *in silico* findings aligned with *in vivo* results, demonstrating that BS (200–400 mg/kg) upregulated *FOS*, *JUN*, *CASP3*, *PI3 K*, and *AKT*, while downregulating *COX2* and reducing hippocampal levels of pro-inflammatory cytokines *TNFα*, *IL-6*, and *IL-1β* in PTZ-treated mice. These molecular changes suggest that BS regulates neuroinflammation in the hippocampus by modulating microglial activation and cytokine production. Importantly, these neurobiological alterations correlated with improvements in locomotor activity and memory performance, further supporting the therapeutic potential of BS in ES/CI.

To further substantiate the beneficial effects of BS on epilepsy, it is noteworthy that BS, a key component of Chaihu Shugan decoction, aligns with its reported efficacy in epilepsy treatment. This TCM formula, traditionally used for chronic gastritis, liver dysfunction, and depression, has been shown to prolong seizure latency in PTZ-induced rat models by enhancing glutamate transporter levels and glutamine synthetase activity (Yu et al. 2015). Moreover, Chaihu Shugan decoction has been successfully combined with antiepileptic drugs (AEDs), including levetiracetam, sodium VLP, and carbamazepine, improving clinical outcomes in refractory epilepsy by reducing seizure frequency (Tao et al. 2023). The pharmacological effects of BS are largely attributed to the antioxidant and anti-inflammatory properties of its active ingredients, particularly bSS and BA (Sun et al. 2020; Jiang and Hao 2020). However, a limitation of this study is the exclusive use of male mice, which restricts insights into potential sex-specific differences. Given that estrogen modulates neuroinflammation and seizure susceptibility (Sato and Woolley 2016), the efficacy of BS may vary in female subjects, warranting further investigation.

Recent research has provided deeper insights into the neuroprotective mechanisms of BS. A comprehensive network pharmacology analysis by Du et al. (2020) identified six bioactive compounds—bSS, BA, kaempferol, paeoniflorin, palbinone, and (+)-catechin—as key contributors to its therapeutic potential, primarily through the regulation of *AKT1*,

IL-6, *CAT*, *TNF*, *CASP3*, and *PTGS* (Du et al. 2020). Building on these findings, an integrated in silico and experimental study by Zeng et al. (2022) demonstrated that *Baishao Luoshi Formula* (BSLSF) could alleviate post-stroke spasticity (PSS) by modulating *AKT1*, *TNF*, *CASP3*, *VEGFA*, and *CREB1*. Their results suggest that *BSLSF* exerts its therapeutic effects primarily through the regulation of synaptic plasticity. Furthermore, *BSLSF* exhibited neuroprotective properties against ischemic stroke, potentially through the modulation of the *CCR5/CREB* signaling pathway, which plays a crucial role in synaptic remodeling and functional recovery (Zeng et al. 2022). Notably, two of these active compounds, bSS and BA, directly target inflammatory pathways. For instance, Sun et al. (2020) demonstrated that bSS mitigates inflammatory responses by modulating the *ERK/p38* and *NF-κB* signaling cascades in *BV2* cells exposed to LPS (Sun et al. 2020). These findings align with evidence that BS attenuates the *IL-17* and *TNF* signaling pathways, significantly reducing hippocampal inflammation. Concurrently, BS upregulates the *PI3 K/AKT* pathway, enhancing neuronal survival. This dual mechanism—suppressing inflammation while promoting cell viability—underpins its ability to restore cognitive function, as demonstrated by improved MWM performance and normalized gene expression in treated mice. The interplay between inflammation and cell survival is pivotal to BS's neuroprotective effects. By reducing pro-inflammatory cytokines such as *TNFα*, *IL-6*, and *IL-1β* through *IL-17* and *TNF* pathway modulation, BS mitigates the neurotoxic environment that drives neuronal death. This reduction in inflammation fosters conditions that enhance cell survival, enabling the *PI3 K/AKT* pathway to activate anti-apoptotic signaling and support synaptic repair. Thus, the anti-inflammatory properties of BS not only limit damage but also amplify its pro-survival effects, leading to improved cognitive outcomes. This integrated action suggests broader therapeutic potential for BS in neuroinflammatory disorders, such as AD, PD, and multiple sclerosis, where inflammation and neuronal loss are hallmarks of progression. By addressing both processes, BS offers a promising approach to managing these complex conditions.

Conclusion and limitations

This study systematically evaluated the therapeutic potential of BS for ES/CI using integrated network pharmacology, molecular docking, and in vivo validation. A total of 6,253 genes associated with epilepsy, cognitive impairment, and inflammation were identified, highlighting the role of inflammatory pathways in ES/CI. In vivo studies confirmed hippocampal neuroinflammation and cognitive impairment in ES-exposed animals, supported by behavioral assessments,

gene expression, and protein analyses. BS exerted its therapeutic effects through bioactive compounds bSS and BA, modulating key targets such as *FOS*, *JUN*, *MAPK1*, *HSP90* *AA1*, and *MTOR*, crucial for neuronal survival and inflammation regulation. The in vivo results aligned with computational predictions, demonstrating BS's ability to regulate inflammatory pathways and improve cognitive function. To advance BS as a potential therapy for ES/CI, chronic efficacy studies are needed to assess its long-term benefits and safety. In addition to the hippocampus, other brain regions, such as the cortex and amygdala, should be considered to gain a more comprehensive understanding of its therapeutic effects. Validating its efficacy in alternative epilepsy models will help establish broader applicability. Investigating combination therapy strategies may also be beneficial, as BS could complement existing antiseizure and neuroprotective treatments like VLP, potentially enhancing efficacy while minimizing side effects. Furthermore, translational research is required to confirm its BBB permeability, assess pharmacokinetics, and conduct phase I/II clinical trials to establish dosing, safety, and efficacy in ES/CI patients. Despite these promising findings, several limitations warrant consideration. The predictive nature of network pharmacology and molecular docking may introduce false positives or overlook minor bioactive compounds. Additionally, while in silico binding affinities suggest strong interactions, they may not fully reflect dynamic in vivo conditions, underscoring the necessity for experimental validation. The study exclusively utilized male mice, limiting the assessment of potential sex-specific differences. Given the influence of hormonal factors on neuroinflammation and seizure susceptibility, future studies should incorporate both sexes. The study design focused on a 21-day PTZ model, effectively evaluating BS's acute effects but not its long-term efficacy, tolerance, or toxicity. Future research should explore chronic treatment effects and alternative epilepsy models to better capture ES/CI complexities. Although bSS and BA were identified as key bioactive compounds, their individual contributions remain unverified, requiring further isolation and evaluation. Safety assessments were limited to general observations without comprehensive toxicity profiling. Additionally, clinical translation is crucial, as BBB permeability must be validated in human models. Pharmacokinetic and pharmacodynamic studies are necessary before clinical trials can be initiated. Ultimately, phase I/II clinical trials will be essential to establish the optimal dosing, safety, and efficacy of BS in ES/CI patients.

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Data availability The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no potential conflicts of interest.

Research involving human participants and/or animals The study was approved by Research Ethics Committee on the Care and Use of Laboratory Animals at Shandong Provincial Third Hospital, Shandong University (No: SDZY220028).

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